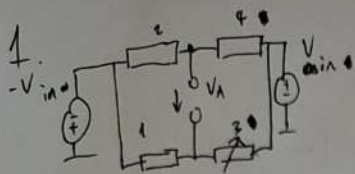


Electronics HW 1



$$1. V_{th} = V_{top} - V_{bottom}$$

$$= \left(-V_{in} + \frac{R_2}{R_2 + R_4} (V_{in} - (-V_{in})) \right) - \left(-V_{in} + \frac{R_1}{R_1 + R_3} (V_{in} - (-V_{in})) \right)$$

$$= V_{in} \frac{R_2 - R_4}{R_2 + R_4} - V_{in} \frac{R_1 - R_3}{R_1 + R_3}$$

$$R_{th} = \frac{R_2 R_4}{R_2 + R_4} + \frac{R_1 R_3}{R_1 + R_3}$$

$$V_A = V_{th} \frac{R_m}{R_{th} + R_m}$$

$$2. R_1 = R_2 = R_3 + R_4 = R$$

$$V_A = V_{th} \frac{R_m}{R_{th} + R_m} = V_{in} \left(\frac{R_2 - R_4}{R_2 + R_4} - \frac{R_1 - R_3}{R_1 + R_3} \right) \frac{R_m}{\frac{R_2 R_4}{R_2 + R_4} + \frac{R_1 R_3}{R_1 + R_3} + R_m}$$

$$= V_{in} \left(0 - \frac{-\Delta R}{2R + \Delta R} \right) \frac{R_m}{\frac{R^2}{2R} + \frac{R^2 + R\Delta R}{2R + \Delta R} + R_m}$$

$$V_{in} = 5V, R = 2k\Omega, R_m = 100k\Omega$$

$$\Delta R = -1k\Omega:$$

$$V_A = 5V \cdot \frac{100000}{4000 + 100000} = \dots \approx -1.639V (3d.p)$$

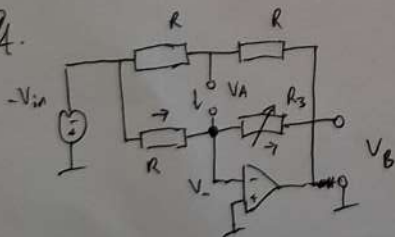
$$\Delta R = 0:$$

$$V_A = 0V$$

$$\Delta R = 100k\Omega:$$

$$V_A \approx 0.978V (3d.p)$$

3.4.



$$4. V_- = 0V \quad (V_+ \text{ grounded, } V_- = V_+ \text{ due to op-amp})$$

$$\frac{V_- - (-V_{in})}{R} + \frac{V_- - V_B}{R_3} = 0 \quad (\text{Kirchhoff current law at inverting node})$$

$$\Rightarrow V_B = V_{in} \frac{R_3}{R}$$

(simple because not dependent on top side)

$$V_A = V_{top} - V_{bottom} = V_{top}$$

$$= -V_{in} + \frac{R}{R+R} (V_{in} - V_{in}) = \frac{-V_{in} + V_B}{2} = \frac{V_{in} \frac{R_3}{R} - V_{in}}{2} = \frac{V_{in} \left(\frac{R_3}{R} - 1 \right)}{2} = \frac{V_{in} \Delta R}{2R}$$

$$4. V_{in} = 5V, R = 2k\Omega, \Delta R = \{-1k\Omega, 0k\Omega, 1k\Omega\}$$

$$\Delta R = -1k\Omega:$$

$$V_A = \frac{5 \cdot (-1000)}{4000} = -1.25V$$

$$\Delta R = 0\Omega:$$

$$V_A = 0V$$

$$\Delta R = 1k\Omega:$$

$$V_A = 1.25V$$

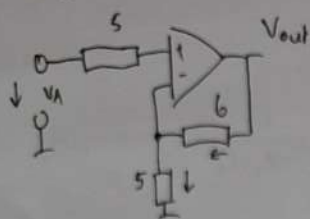
(VA)

It is now symmetric around $\Delta R = 0$ and linear to ΔR unlike before.

$\Rightarrow$ easier to work with

$\Rightarrow$ change in R_3 can use to change V_A easily

5.



$$I_+ = I_- = 0 \text{ A} \quad (\text{op-amp})$$

$$\Rightarrow R_5 \cdot 0 \text{ A} \Rightarrow V_{R_5} = I R_5 = 0 \text{ V}$$

$$V_- = V_+ = V_A - V_{R_5} = V_A$$

$$\frac{V_{out} - V_-}{R_6} + \frac{V_- - 0}{R_5} = 0 \quad (\text{Kirchoff current law at } V_-)$$

$$\Rightarrow V_{out} = V_A \left(1 + \frac{R_6}{R_5} \right)$$